Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/24/2006
PI Name:	Heinzen, Daniel J. Ph.D.		
Project Title:	Search for Time Reversal Symmetry Violation with Laser-Cooled Atoms		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Austin	**State:**	TX
Zip Code:	78712-0264	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
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Flight Program:			
Flight Assignment:	NOTE: is on NCE until 4/30/07 (per info fr N. Hall 9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2908		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	Our objective is to search for time-reversal symmetry (T) violation with ultracold atoms. In order to do so, we will search for a T-violating electric dipole moment (EDM) of trapped Cs atoms. The apparatus will consist of two pairs of electric field plates inside a high vacuum chamber. The entire chamber will be placed in magnetic shields. Laser cooled Cs atoms will be loaded into a far-off resonance optical dipole force trap. We will probe the Zeeman resonances of the trapped Cs atoms, and look for the first order Stark effect that would result from an interaction between the electric field and the atomic electric dipole moment. Our objective is to improve the current bound on the electric dipole moment of the Cs atom by a factor of 100. The search for time reversal symmetry (T) violation is one of the most intriguing problems in physics. Such T-violation must have occurred in the early universe, because otherwise no matter would have remained after its annihilation with anti-matter. However, almost nothing is known about T-violation. If our experiment succeeds in observing T-violation, it will shed light on this most important scientific puzzle. Even if we observe no T-violating effect at a more sensitive level, we will still provide important constraints on possible models of T-violation. This scientific question is of a sufficiently important and basic nature that it is of interest to a very general audience.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)